Wi-Fi 7 Link Quality in Indoor and Outdoor Environments

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Supervisor: Professor Nurul Sarkar · Final Technical Report, Parts A & B · 2026

1 Introduction & Objectives

Wi-Fi 7 is fast on paper. We tested it in real places.

The problem. Wi-Fi 7 (IEEE 802.11be) [1] promises higher speeds and Multi-Link Operation (MLO), which combines the 2.4, 5 and 6 GHz bands. But real speed depends on distance, walls, floors and the band used. We measured what actually happens.

Objectives

1. Measure how walls and floors weaken the signal indoors
2. Measure how distance affects speed outdoors (10–100 m)
3. Test whether MLO beats the best single band

2 Methodology

Two tests, one question: what weakens the link?

Same room · 1 wall · 2 walls · 3 walls · Floor 2 · Floor 3

Part A · Indoor, AUT building. Access point on Floor 1. At each of 6 points we logged signal strength (RSSI, dBm) and quality (SNR, dB) on 2.4, 5 and 6 GHz with NetSpot Pro [5].

AP · 10 · 20 · 30 · 40 · 50 · 60 · 70 · 80 · 90 · 100

Part B · Outdoor, William Green Domain. Football pitch, Highland Park. At 10 points (10–100 m) we ran iPerf3 [4] speed tests and logged RSSI/SNR for each band and MLO.

3 Results

Distance and obstacles decide which band wins.

698 Mbps
6 GHz download at 10 m, fastest single band

No link
6 GHz at 100 m
0 Mbps, −93 dBm

55 Mbps
MLO download at 100 m, still connected

+76.5%
MLO gain at 80 m, vs best single band

A Indoor · AUT building

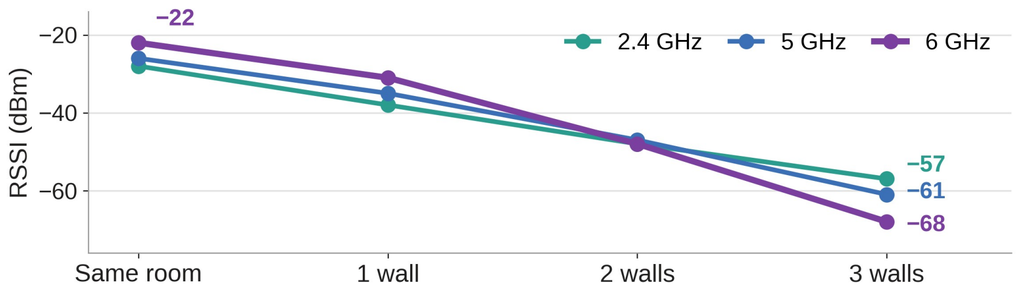


Figure 1. *RSSI (dBm; closer to 0 = stronger) falls with every wall. 6 GHz is strongest in the same room but weakest behind 3 walls.*

Band	Floor 2	Floor 3
2.4 GHz	−52 dBm · SNR 16 Fair	−67 dBm · SNR 5 Weak
5 GHz	−55 dBm · SNR 13 Weak	−71 dBm · SNR 2 Very weak
6 GHz	−62 dBm · SNR 6 Weak	−84 dBm · SNR 0.9 No reliable signal

Table 1. *Client on Floor 2 or 3, access point on Floor 1. On Floor 3, 6 GHz had no reliable signal.*

B Outdoor · William Green Domain

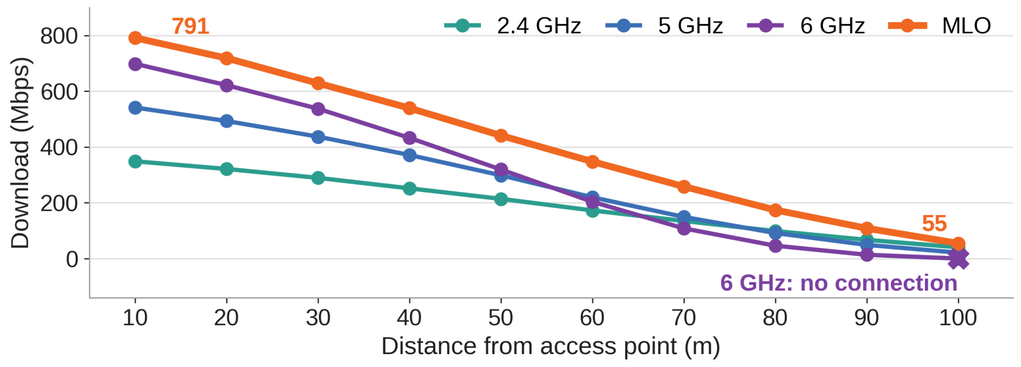


Figure 2. *Download speed vs distance. 6 GHz wins up close but drops to zero at 100 m; MLO is fastest at every point.*

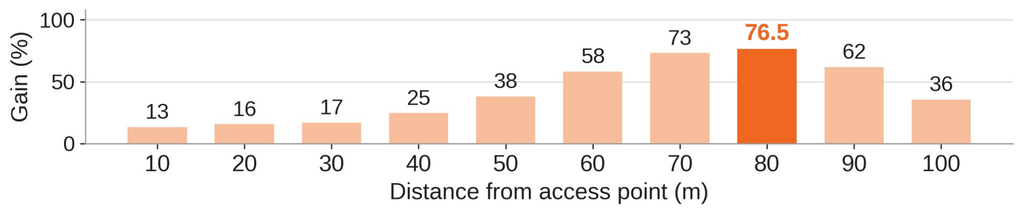


Figure 3. *MLO gain = how much faster MLO is than the best single band: +13% at 10 m, peak +76.5% at 80 m.*

4 Conclusions & Future Work

MLO kept the link alive where single bands failed.

What we found

- Indoors, every wall and floor weakens the signal; 6 GHz is best in the same room but unusable on Floor 3.
- Outdoors, 6 GHz is fastest at 10 m but drops out at 100 m; 2.4 GHz is slower but still connects.
- MLO beat the best single band at all 10 points (+13% to +76.5%).

Limitations

- One site, one AP/client pair
- No repeat-run averages reported
- Raw logs not yet attached
- MLO alone not proven as the cause

Next steps

- Repeat runs, report averages
- Attach raw iPerf3/NetSpot logs
- Compare centre vs edge AP
- Test other sites and devices

5 Relevance to Industry

Plan Wi-Fi 7 with measured data, not peak speeds.

- **Who needs this?** Sports clubs and outdoor venues (event operations, staff communications, media) and campus or office IT teams. One access point can leave dead zones: in our tests 6 GHz had no link at 100 m or on Floor 3.
- **What could they use?** Band-by-distance data and a repeatable survey method (NetSpot Pro + iPerf3), field-tested at one site.
- **Why better than now?** Versus the best single band, MLO gave up to 76.5% more speed (174 vs 99 Mbps at 80 m).
- **What would adoption take?** Wi-Fi 7 APs and clients that support MLO, a site survey per venue, and compliance with NZ 6 GHz rules (RSM). Cost not assessed; next step is centre vs edge AP trials.

One-line impact: **“This work lets venue and campus IT teams plan Wi-Fi 7 from measured data: MLO gave up to 76.5% more speed than the best single band.”**

Author contributions: Santiago Viscarra: indoor survey, outdoor measurements, analysis and poster. Supervised by Professor Nurul Sarkar.

References: [1] IEEE Std 802.11be-2024. [2] Sohal et al. (2026), AUT report on noise and interference in Wi-Fi 7. [3] Roberts et al. (2026), AUT report on indoor radio propagation. [4] iPerf3, iperf.fr. [5] NetSpot Pro, netspotapp.com.